

## Submission LINALG1-MID-SUB08

### Question LINALG1-MID-Q01

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visible work:

$$\det = 2(3) - 4(2) = -2 \dots ?$$

last line is hard to read; maybe  $x=1$ ,  $y=2$

### Question LINALG1-MID-Q02

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visible work:

$$\det(A) = 6 \cdot 4 - 2 \cdot 2 = 20 \dots ?$$

last line is hard to read; maybe  $\det=20$ ; invertible

### Question LINALG1-MID-Q03

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visible work:

$$T(1, -1) = (1 - 2, 3 + 1) = (-1, 4) \dots ?$$

last line is hard to read; maybe  $T(1, -1) = (-1, 4)$ ; matrix  $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ ; linear

### Question LINALG1-MID-Q04

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visible work:

$$\text{Set } c \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (0, 0) \dots ?$$

last line is hard to read; maybe only trivial combination;  $\{(1, 0), (0, 1)\}$  is linearly independent

### Question LINALG1-MID-Q05

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visible work:

$\text{Null}(A)$  is the set of vectors sent to the zero vector.... ?

last line is hard to read; maybe the vectors mapped to zero; exactly the solution space of  $Ax=0$

### Question LINALG1-MID-Q06

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visible work:

$$\det = 5(5) - 2(2) = 21 \dots ?$$

last line is hard to read; maybe  $x=1$ ,  $y=2$

### Question LINALG1-MID-Q07

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visible work:

$$\det(A) = 2 \cdot 2 - 2 \cdot 2 = 0 \dots ?$$

last line is hard to read; maybe  $\det=0$ ; not invertible

Question LINALG1-MID-Q08

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visible work:

$$T(1, -1) = (1 - 2, 3 + 1) = (-1, 4) \dots ?$$

last line is hard to read; maybe  $T(1, -1) = (-1, 4)$ ; matrix  $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ ; linear